

Supermarket Sales Analysis

Data Analytics Internship Project Report

1. Problem Statement

The objective of this project is to analyze supermarket sales data and identify useful information about products, branches, categories, customers, payment methods, and customer ratings.

2. Dataset

The dataset contains 500 sales transactions. It includes product, branch, city, customer type, quantity, unit price, payment method, rating, and sales-related information.

3. Methodology

The project loads the CSV dataset, checks missing and incorrect values, calculates sales using $\text{Quantity} \times \text{Unit Price}$ where necessary, groups data using totals, counts, and averages, creates charts, and interprets the findings for business decisions.

4. Key Findings

Cheese generated the highest product sales with ₹27,906.30. Branch C (Mumbai) performed best with sales of ₹72,469.45. Beverages had the highest category sales with ₹56,108.24. UPI was the most-used payment method with 127 transactions. The average Member transaction was ₹483.14, while Normal customers averaged ₹497.07. The average customer rating was 3.99 out of 5.

5. Business Recommendations

The supermarket can maintain higher stock levels for high-selling products and categories, study the practices contributing to Branch C's performance, continue supporting UPI payments, and improve customer service to increase ratings. Customer spending patterns can also support targeted membership offers.

6. Conclusion

This project demonstrates how supermarket sales data can be transformed into practical insights. The analysis supports better understanding of sales performance, customer behavior, payment preferences, and opportunities for business improvement.

Project Summary

Metric	Result
Highest-selling product	Cheese – ₹27,906.30
Best-performing branch	Branch C (Mumbai) – ₹72,469.45
Highest-selling category	Beverages – ₹56,108.24
Most-used payment method	UPI – 127 transactions
Average Member transaction	₹483.14
Average Normal transaction	₹497.07
Average customer rating	3.99 / 5

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For internship submission